Quantification: 701_z36c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	3
interior lacuna count	3
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 3)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	883.667	804	491.368	px^2
major_axis_length_px	lacuna shape	42.365	41.5213	11.3585	px
minor_axis_length_px	lacuna shape	28.8076	33.4623	9.1056	px
aspect_ratio	lacuna shape	1.5067	1.5622	0.243	unitless
eccentricity	lacuna shape	0.7244	0.7683	0.1168	unitless
solidity	lacuna shape	0.8359	0.8918	0.123	unitless
roots_count	headline	7.6667	8	0.5774	count
ring_length_r30_px	headline	402.333	440	92.4464	px
ring_length_r60_px	reported	1110.67	1210	176.398	px
owned_length_px	ownership-dependent	2182.17	1329.13	1514.13	px
edge_count	ownership-dependent	162.333	91	129.662	count
mean_edge_length_px	ownership-dependent	14.1749	14.6058	1.4123	px
perimeter_px	normalised	160.123	158.267	66.6027	px
ring_area_r30_px2	normalised	6434	6849	1035.85	px^2
ring_area_r60_px2	normalised	18434.3	19116	2030.22	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.0620188	0.064243	0.0047932	px^-1
ring_density_r60	normalised	0.060032	0.0606409	0.00360885	px^-1
roots_per_100px_perimeter	normalised	5.3263	5.0547	1.9619	per 100 px
ring_attached_length_r30_px	network v2	318.667	313	44.7698	px
ring_attached_length_r60_px	network v2	660.333	590	192.396	px
ring_length_w_r30_px	network v2	461.623	504.375	101.213	px
ring_length_w_r60_px	network v2	1252.59	1365.13	200.507	px
sholl_crossings_r10	network v2	10.3333	11	4.0415	count
sholl_crossings_r20	network v2	8.6667	9	0.5774	count
sholl_crossings_r30	network v2	14.6667	18	5.7735	count
loop_count	loops, report only	9.3333	4	10.116	count
loop_edge_share	loops, report only	0.2285	0.1758	0.0992	share
ring_width_mean_r30_px	width	5.7781	5.7803	1.0654	px

Per field
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measure	value	units
analysed_area_px2	767286	px^2
total_skeleton_length_px	43384	px
canalicular_length_density_per_px	0.0565421	px^-1
junction_count	1304	count
junction_density_per_px2	0.0016995	px^-2
lacuna_count	3	count

lacunae_per_px2	3.91e-06	px^-2
skeleton_component_count	715	count
mean_component_length_px	60.6769	px
median_component_length_px	12	px
field_length_density_w_per_px	0.0635222	px^-1
field_density_without_flagged_per_px	0.0401463	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	64	px
unattached_component_count	5	count
canalicular_length_density_attached_per_px	0.0565421	px^-1
field_length_density_with_unattached_per_px	0.0566256	px^-1
excluded_area_px2	278639	px^2
analysed_field_area_px2	769937	px^2
loop_count	175	count
loop_edge_share	0.1981	share
edge_count	4215	count

Canalicular width, the new measure

measure	value	units
width_mean_px	5.6859	px
width_median_px	5.6569	px
width_p10_px	2	px
width_p90_px	8.9443	px
width_pixels_used	36059	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	804	41.5213	33.4623	1.2408	0.592	0.6949	-0.5169	176.293	670.124	False
2	437	31.4519	18.3155	1.7172	0.8129	0.8918	1.4887	285.545	929.915	False
3	1410	54.1218	34.645	1.5622	0.7683	0.921	-1.5551	413.697	828.063	False

Network per lacuna

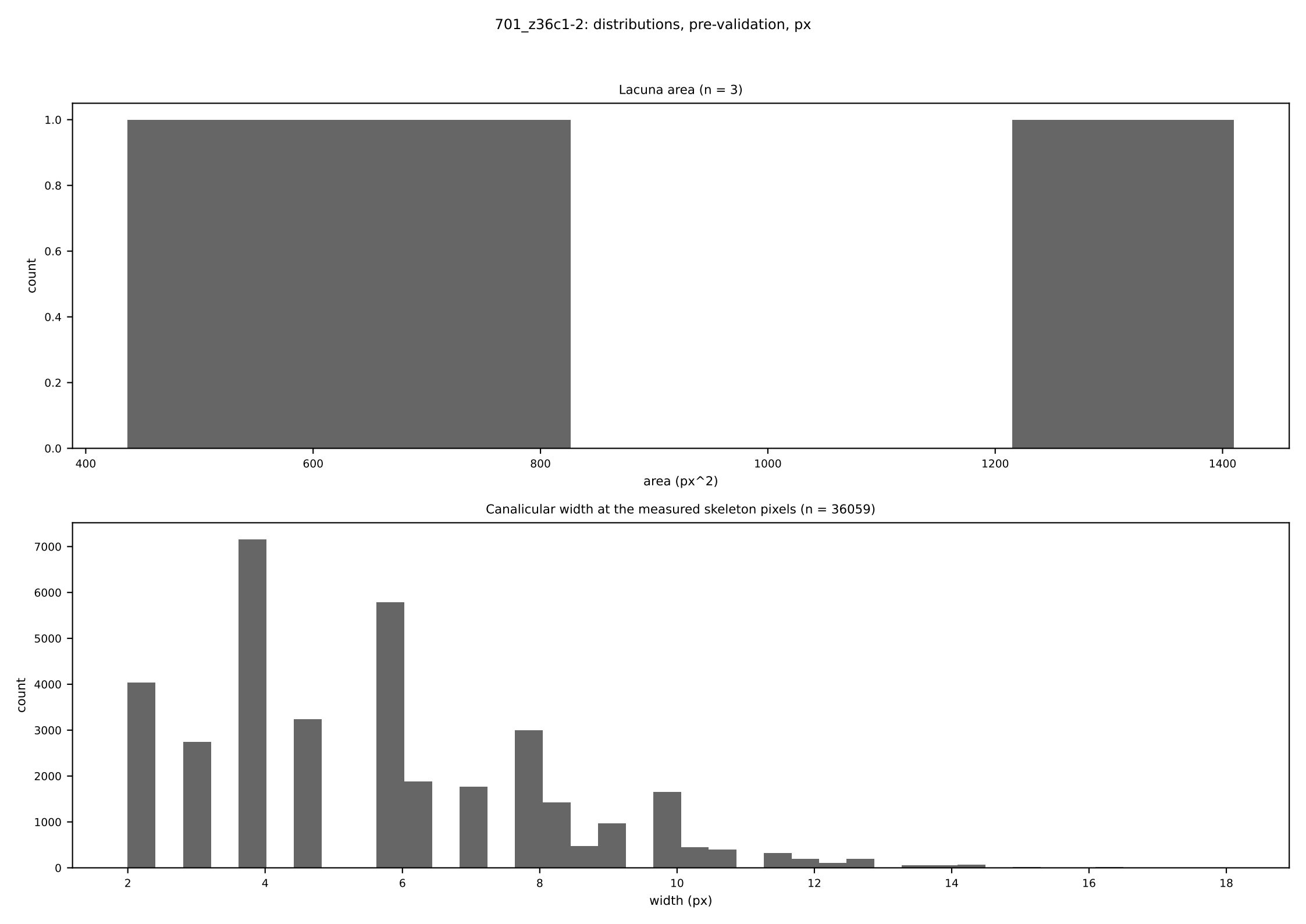
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	8	440	1210	3930.36	312	12.5973	5.7803
2	7	297	907	1329.13	91	14.6058	6.8423
3	8	470	1215	1287	84	15.3215	4.7116

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	227.635	6849	19116	1	1	0.064243	0.0632978	3.5144
2	94.468	5255	16151	1	1	0.0565176	0.0561575	7.4099
3	158.267	7198	20036	1	1	0.0652959	0.0606409	5.0547

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	366	878	504.375	1371.54	11	8	18
2	277	513	346.049	1021.09	6	9	8
3	313	590	534.446	1365.13	14	9	18


701_z36c1-2: distributions, pre-validation, px
Lacuna area (n = 3)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 36059)
count
width (px)